

PRODUCT DESCRIPTION

Revival is a single-component, waterborne rust converter and primer for ferrous metal. Applied directly onto rusted steel, the acidic conversion system in the product reacts with the existing iron oxide and converts it into a stable, insoluble compound that is bound to the substrate and no longer able to propagate; the polymer binder films over the converted layer and seals it against moisture and oxygen. The visible indicator is a change of colour: the orange-brown rust darkens. The resulting film functions as a primer, so a finishing paint can be applied directly without a separate primer coat. Revival is intended for large-area and awkward-access work — structural steelwork, plant pipework, roof structures, tanks, fences and vehicle chassis — where complete removal of rust by abrasion is impractical in terms of time and labour cost. It is a primer, not a finish: a topcoat is required.

KEY DATA

Type	Single-component waterborne rust converter and primer for ferrous metal
Binder system	Waterborne polymer dispersion (composition proprietary)
Active principle	Mildly acidic rust-conversion system (composition proprietary)
Appearance — liquid	Milky white to off-white liquid; low odour
Appearance — dry film	Dark film. The final shade varies with the type, age and condition of the rust and ranges from black through dark brown to grey; an even black finish across a whole component should not be expected
Volatile organic content	Less than 1 % by weight
Heavy metals	Contains no lead and no heavy metals
Flammability	Not flammable — waterborne system
Suitable substrates	Rusted or bare ferrous metal (steel, cast iron, wrought iron)
Unsuitable substrates	Painted or coated surfaces; galvanised, zinc-coated or zinc-sprayed steel; aluminium; stainless steel; other non-ferrous metals
Surface preparation	Remove loose, flaking and blistered scale by hammer, wire brush or power tool; retain tightly-adhered rust. Degrease and remove oil, grease and dirt. The surface must be dry
Application	Brush or roller. Apply generously and work the liquid into pits, welds, bolt heads and corners; this is a penetrating treatment, not a decorative thin coat
Number of coats	One coat on light rust; two coats on heavy, thick or long-weathered rust
Colour change	Typically visible within 15–30 minutes
Recoat interval	2–4 hours between coats
Overcoating	Minimum 24 hours before the finishing paint; allow up to 48 hours in humid conditions or at low temperature. A topcoat is required — see Application Notes
Application conditions	Above 10 °C, dry weather; do not apply if rain is expected within 4–6 hours, onto a wet or frosted surface, or onto steel that is hot in direct sun
Thinning	Do not thin. Dilution weakens the conversion reaction in direct proportion
Coverage	Not quoted as a fixed figure. Consumption depends strongly on the roughness and porosity of the rust profile and can differ by several times between a rough corroded surface and smooth bare steel
Cleaning of equipment	Water, immediately while the product is still wet
Packaging	500 g · 1 kg · 5 kg · 15 kg

MECHANISM

Rust is not a single substance; it is a layered mixture of iron oxides and hydroxides that remains chemically active and continues to grow as long as moisture and oxygen reach it. Conventional anti-corrosive paint applied over rust encapsulates that active layer without changing it, and the reaction continues underneath until the film is lifted and fails from below. Revival works differently: the acidic component reacts with the active oxide and converts it into a stable, insoluble compound anchored to the sound metal, while the polymer binder forms a continuous film over the converted layer and excludes moisture and oxygen. Two consequences follow directly from this mechanism. First, the liquid must physically reach the rust, which is why loose scale and blistered plate must be removed — material that the liquid cannot penetrate will continue to corrode beneath a surface that has apparently changed colour. Second, the product requires rust in order to work; on a clean painted surface there is nothing for it to react with.

NOTE ON VALUES

The times given are typical at approximately 30 °C and moderate humidity and will lengthen in cool, humid or poorly ventilated conditions. No coverage figure is quoted because consumption is governed by the rust profile rather than by the geometric area. The performance obtained is determined principally by the standard of surface preparation and by the protective quality of the topcoat selected. Figures in this document are typical guidance only and are not a specification.

APPLICATION NOTES

- Apply to rusted or bare ferrous metal only. Do not apply over existing paint or coatings: the product is acidic and will attack the coating, and there is no rust present for it to convert. Mask adjacent painted areas before starting and wipe any splashes off paintwork immediately while still wet, using a cloth damped with clean water.
- Do not apply to galvanised, zinc-coated or zinc-sprayed steel, to aluminium, or to other non-ferrous metals. An acidic system will attack a sacrificial zinc layer and remove the protection it provides.
- Remove loose, flaking and blistered scale first, until only tightly-adhered rust remains. It is not necessary, and not intended, to abrade back to bright metal. Where corrosion has perforated the section or reduced it until it deflects under hand pressure, conversion alone is not an adequate repair and the section must be reinstated or replaced.
- Stir the contents thoroughly before use; the product can separate on standing. Do not dilute.
- Apply generously with brush or roller and work the liquid into pits, weld seams, bolt heads and internal corners. A thin, evenly spread film applied as if it were paint will not penetrate a heavy rust layer.
- Judge the result by the change in colour: the treated surface must be clearly darker than before. An even black across the whole component is not required and will not always occur. Any area that remains bright orange has not been reached by the liquid and should be treated again.
- On heavy or long-weathered rust apply a second coat once the first is dry to the touch, after 2–4 hours.
- Revival is a primer, not a finish. The converted film is not formulated to stand exposed to sunlight and weather on its own and gives only limited protection if left uncoated. Always apply a finishing paint over it once the film is fully dry; the topcoat is the layer that carries the long-term protection. No separate primer is required beneath that topcoat.
- Because the product is waterborne, allow the water to evaporate completely before overcoating. Painting over a film that is not fully dry traps moisture beneath the finish and the problem returns within months.
- Clean brushes, rollers and trays with water immediately after use. Once the product has dried on equipment it is very difficult to remove.

HEALTH & SAFETY

The product is a mildly acidic, waterborne liquid. It is not flammable and has a low odour and a low volatile organic content, but it should be treated as an irritant: on the basis of its composition it is expected to cause eye irritation, and skin irritation on prolonged or repeated contact. Full composition is withheld as proprietary; a regulatory Safety Data Sheet is available to purchasers on request.

- Personal protection: wear chemical-resistant gloves (nitrile or rubber) and eye protection. Cover exposed skin when working overhead or on structures where run-off is likely. Work with adequate ventilation.
- Eye contact: rinse immediately with clean water for at least 15 minutes, holding the eyelids apart, and obtain medical advice.
- Skin contact: wash with soap and plenty of water. Remove contaminated clothing and wash it before re-use. Obtain medical advice if irritation persists.
- Ingestion: do not induce vomiting. Rinse the mouth and drink water. Obtain medical advice immediately and show this document.
- Inhalation: the product is low-odour and low-VOC and inhalation is not the principal route of exposure. If mist is generated in a confined space, move the affected person to fresh air.
- Incompatibility: being acidic, the product must not be mixed with alkalis, cement or cementitious materials, or with hypochlorite bleach. Do not decant or store it in unlined steel, zinc or galvanised containers, which it will corrode.
- Fire: not flammable. In a surrounding fire, use any extinguishing medium suitable for the materials involved.
- Spillage: contain with inert absorbent material such as sand or earth, collect into a labelled closed container and dispose of it as chemical waste. Do not allow the product to enter drains, soil or watercourses. Wash the area with plenty of water.
- Disposal: dispose of residues and containers in accordance with local regulations. Do not empty into drains. Do not re-use empty containers for food, drink or animal feed.
- Keep out of the reach of children and animals. Keep the container closed and clearly labelled.

STORAGE

Store in the original, tightly closed container in a cool, dry, well-ventilated place, out of direct sunlight. The product is a waterborne dispersion and must be protected from freezing; a container that has frozen should be discarded, as the dispersion does not recover. Reseal immediately after use and stir before each use. Shelf life 12 months from delivery when stored as described.